

Zhidan Li

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Latest Update: September 12, 2026

Experience	• <i>Visiting Postgraduate Research Student</i> 2022.08 - 2022.10 School of Informatics, University of Edinburgh Supervisor: <i>Heng Guo</i>
Education	• <i>Doctor of Philosophy</i> in Computer Science 2023.09 - current John Hopcroft Center, Shanghai Jiao Tong University Advisor: <i>Kuan Yang</i> • <i>Bachelor of Engineering</i> in Computer Science 2019.09 - 2023.06 ACM Honor Class, Shanghai Jiao Tong University
Research Interests	• Approximate counting and sampling in theoretical computer science. • Counting and sampling problems on random structures. • Algorithmic implications from statistical physics. • High-dimensional probability and geometry.
Research Articles	• Spatial mixing and deterministic approximate counting of multi-spin systems beyond bounded degree graphs Zhidan Li and Kuan Yang Preprint. • An FPRAS for antiferromagnetic Ising models on random regular bipartite graphs Zhidan Li and Kuan Yang Preprint. • Counting and sampling anti-ferromagnetic Potts models on random regular bipartite graphs in the non-uniqueness regime Zhidan Li, Siyu Liu and Kuan Yang Preprint. • FPTAS for Holant problems with log-concave signatures Kun He, Zhidan Li, Guoliang Qiu and Chihao Zhang In <i>Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2025)</i>.
Teaching	• In Shanghai Jiao Tong University: – Serve as a <i>Teaching Assistant</i>: * CS3905: <i>Algorithm Design and Analysis</i> Spring 2023 Instructor: <i>Yuhao Zhang</i> * CS2910: <i>Optimization</i> Summer 2023 Speaker: <i>Yunwei Ren</i> * MATH3806: <i>Optimization Methods</i> Autumn 2023 Instructors: <i>Bo Jiang</i> and <i>Kuan Yang</i>

* CS0901: *Combinatorics*
Instructor: *Kuan Yang*

Autumn 2024, 2025

Seminars

- In Shanghai Jiao Tong University:

- Serve as a *Speaker*:

- * Reading Seminar on *Localization Schemes* Spring 2023
Organiser: *Chihao Zhang*
 - * Reading Seminar on *Introductions to Random Graphs* Spring 2024
Organiser: *Kuan Yang*